

NDU Project - Staging Branch Report

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1. Executive Summary

This report summarizes the uncommitted changes on the staging branch of the NDU Project, a Flutter/Dart project management application. The changes span 19 modified files and 8 new untracked files, totaling approximately 541 insertions and 331 deletions across the codebase.

Key areas of work include: Firebase initialization reliability improvements, account lockout security hardening, null-safety guard rails for schedule screens, methodology synchronization across WBS/Schedule/Cost modules, UI refinements (collapsible navigation persistence, button sizing), cost estimate import improvements, and the addition of a local AI assistant Modelfile.

2. Files Changed (Uncommitted)

Modified Files (19)

- lib/main.dart
- lib/services/security_services.dart
- lib/screens/sign_in_screen.dart
- lib/screens/project_framework_next_screen.dart
- lib/screens/cost_estimate_screen.dart
- lib/widgets/section_navigator.dart
- lib/widgets/launch_phase_navigation.dart
- lib/utils/project_data_helper.dart
- lib/cost_estimate/providers/cost_estimate_provider.dart
- lib/cost_estimate/screens/cost_estimate_module_screen.dart
- lib/schedule/providers/schedule_provider.dart
- lib/schedule/screens/builder_screen.dart
- lib/schedule/screens/gantt_screen.dart
- lib/screens/list_view_screen.dart
- lib/schedule/screens/schedule_module_screen.dart
- lib/wbs/providers/wbs_provider.dart
- lib/wbs/screens/wbs_builder_screen.dart
- lib/wbs/screens/wbs_module_screen.dart
- lib/project_controls/screens/project_controls_screen.dart

New/Untracked Files (8)

- test/account_lockout_test.dart
- test/cost_estimate_initiation_import_test.dart
- test/schedule_builder_null_schedule_test.dart
- test/schedule_module_collapsed_content_test.dart
- test/schedule_navigator_collapse_test.dart
- test/wbs_methodology_sync_test.dart
- ollama/ndu-assistant.Modelfile
- lib/screens/image.png

3. Detailed Changes

3.1 Firebase Initialization Reliability (lib/main.dart)

The Firebase initialization flow has been refactored into a dedicated `initializeFirebase()` function that returns a boolean success indicator.

- Added `FirebaseBootstrap` class to track initialization state across the app
- Made initialization idempotent (checks `Firebase.apps.isEmpty` before re-initializing)
- Centralized the timeout (10s) and error handling in one place
- Enabled retry capability: the UI can call `initializeFirebase()` again on failure
- Moved all post-init setup (API key, image cache tuning, warm-up, session manager) after Firebase is ready
- Image cache reduced from 100MB/1000 images to 50MB/500 images for web optimization

3.2 Account Lockout Security Hardening

Both `security_services.dart` and `sign_in_screen.dart` have been updated to prevent infrastructure errors from locking users out of their accounts.

- Added `credentialErrorCodes` set: `{invalid-credential, wrong-password, user-not-found}`
- New `isCredentialErrorCode()` helper to classify Firebase error codes
- New `recordFailedAttemptForCode(errorCode)` method: only records failures for actual bad credentials
- Network errors, Firebase misconfiguration, popup blocks, and server throttling no longer count toward lockout
- Generic catch block no longer records failed attempts (logs only)
- Remaining-attempts message only shown for credential errors

3.3 Null-Safety Guard Rails for Schedule Screens

All three schedule sub-screens (Builder, Gantt, ListView) previously force-unwrapped `provider.schedule!` which could crash during async loading.

- `builder_screen.dart`: Changed `provider.schedule!` to nullable check with loading spinner
- `gantt_screen.dart`: Same null guard added
- `list_view_screen.dart`: Same null guard added
- All three show a centered `CircularProgressIndicator` with "Loading schedule..." text

3.4 Methodology Synchronization Across Modules

When the user changes the project methodology in Project Details (Waterfall/Agile/Hybrid), the WBS, Schedule, and Cost modules now stay in sync.

- Schedule Provider: New `syncDeliveryModel(String)` method updates `basis.deliveryModel`
- Schedule Module Screen: Calls `syncDeliveryModel` on build to match project methodology
- WBS Provider: New `frameworkForMethodology()` static helper maps methodology to WBS framework
- WBS Provider: New `syncMethodology(ProjectMethodology)` updates document-level methodology and framework
- Project Data Helper: New `deliveryModelForMethodology()` converts `ProjectMethodology` to AGILE/WATERFALL/HYBRID string

3.5 Cost Estimate Improvements

Cost Estimate Provider: Wrapped `FirebaseAuth.instance.currentUser` in try-catch for test/Firebase-not-initialized safety

- Same safety wrap in `_syncAuthenticatedOwner()`
- Import from project initiation now recomputes totals (was leaving dashboard KPI at \$0)
- Cost Estimate Module Screen: Auto-populates cost lines from Project Details initiation phase when estimate is empty

- Calls `importFromProjectCostEstimateItems()` and `ensureSeededFromProjectData()`
- Cost Estimate Screen: Removed redundant `_TopUtilityBar` widget (back/forward navigation)
- Cost Estimate Screen: Removed `_UserChip` widget (duplicated from header bar)

3.6 UI Refinements

- Section Navigator: Added collapse state persistence via `SharedPreferences`
- New `collapseStorageKey` parameter for per-screen persistence
- Static in-memory cache (`_collapsedCache`) for synchronous initial state
- Launch Phase Navigation: Reduced button padding and icon sizes for compactness
- Added `tapTargetSize`: `MaterialTapTargetSize.shrinkWrap` for better tap targets
- Project Framework Next Screen: Goal card header changed from Row to Column layout
- Goal card title field now allows 3 lines (was 1)

- Project Controls Screen: SectionNavigator made collapsible with initiallyCollapsed: true
- Schedule/Cost Module Screens: SectionNavigator defaults to collapsed

4. New Local AI Assistant (ollama/ndu-assistant.Modelfile)

A new Ollama Modelfile has been added to create a local NDU Assistant model based on qwen2.5-coder:14b.

- System prompt covers the full NDU Project architecture (Flutter + Firebase)
- Knows all key modules: WBS, Schedule, Cost, Risk, Quality, Charter
- Follows existing code conventions (const constructors, Provider pattern, TreasuryTokens)
- Configured for low-temperature (0.3) code generation
- Context window: 16384 tokens

5. New Test Files

Six new test files have been added covering the changes in this batch:

- test/account_lockout_test.dart: Tests credential-error-only lockout logic
- test/cost_estimate_initiation_import_test.dart: Tests auto-import from project initiation
- test/schedule_builder_null_schedule_test.dart: Tests null schedule loading state
- test/schedule_module_collapsed_content_test.dart: Tests collapsed section behavior
- test/schedule_navigator_collapse_test.dart: Tests SectionNavigator collapse persistence
- test/wbs_methodology_sync_test.dart: Tests WBS methodology sync with project details

6. Summary Statistics

Total modified files: 19

Total new/untracked files: 8

Lines added: ~541

Lines removed: ~331

Net change: +210 lines

New test files: 6

Modules affected: Firebase Init, Security, Schedule, WBS, Cost Estimate, UI Widgets

Report generated by Codebuff